Finance Theory for AI Applications

BUS 696: GENERATIVE AI IN FINANCE

WEEK 2

Professor Jonathan Hersh
Chapman University, Argyros School of Business
Spring 2026

Course Roadmap

01

Foundation Review

Quick recap of Week 1 essentials: returns, risk metrics, Sharpe ratio, and technical indicators

02

Market Mechanics

Understanding price formation through the lens of marginal buyers and sellers

03

Risk Distribution

Examining whether stock returns follow normal distributions and implications for AI models

04

CAPM Framework

Capital Asset Pricing Model: predicting expected returns based on systematic risk

05

Options Primer

Introduction to derivatives and asymmetric risk-reward structures

06

Portfolio Theory

Preview of diversification benefits and mean-variance optimization

07

Hands-On Lab

Applying concepts through Python-based financial analysis

Week 1 Foundation: Core Concepts

Return Calculation

Percentage gain or loss expressed as:

$$\frac{P_{today} - P_{yesterday}}{P_{yesterday}}$$

Volatility Measurement

Standard deviation of returns quantifies price variability and investment risk

Key Performance Metrics

Sharpe Ratio: Risk-adjusted return metric calculated as (Return – Risk-free rate) / Volatility. Higher values indicate better return per unit of risk taken.

Moving Averages: Smoothed price trends used to identify momentum and potential trading signals in technical analysis.

yfinance Library: Our Python toolkit for accessing historical market data and conducting quantitative research.

These fundamental concepts form the analytical foundation we'll build upon throughout today's session, connecting classical finance theory to modern AI applications.

The Mechanics of Price Formation

Stock prices don't exist in a vacuum—they emerge from a continuous auction process where buyers and sellers negotiate in real-time. Understanding this microstructure is essential for developing effective AI trading strategies.

Bid Price

The highest price any buyer is currently willing to pay. This represents demand—the most aggressive buyer in the market at this instant.

Ask Price

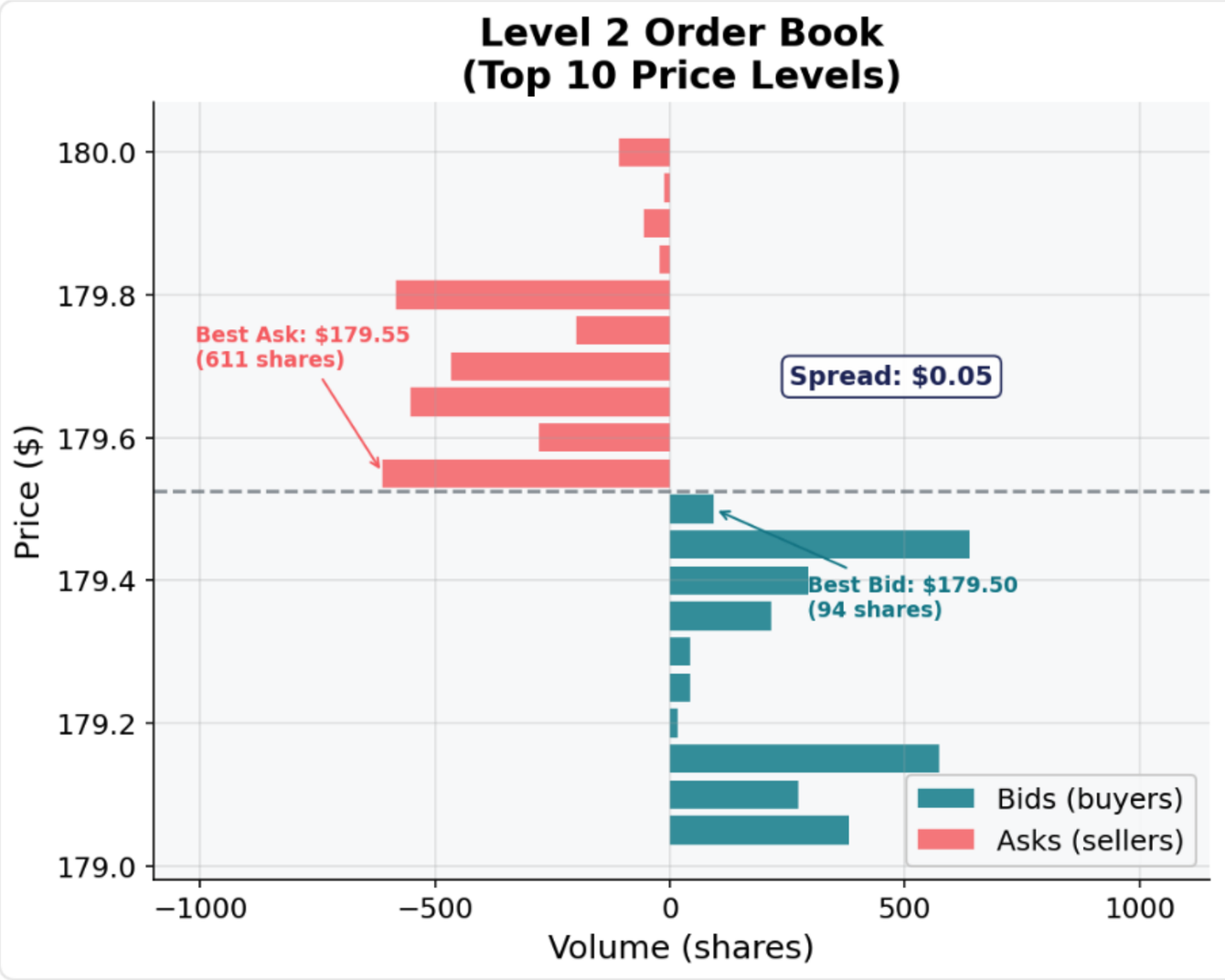
The lowest price any seller is willing to accept. This represents supply—the most motivated seller currently in the market.

Bid-Ask Spread

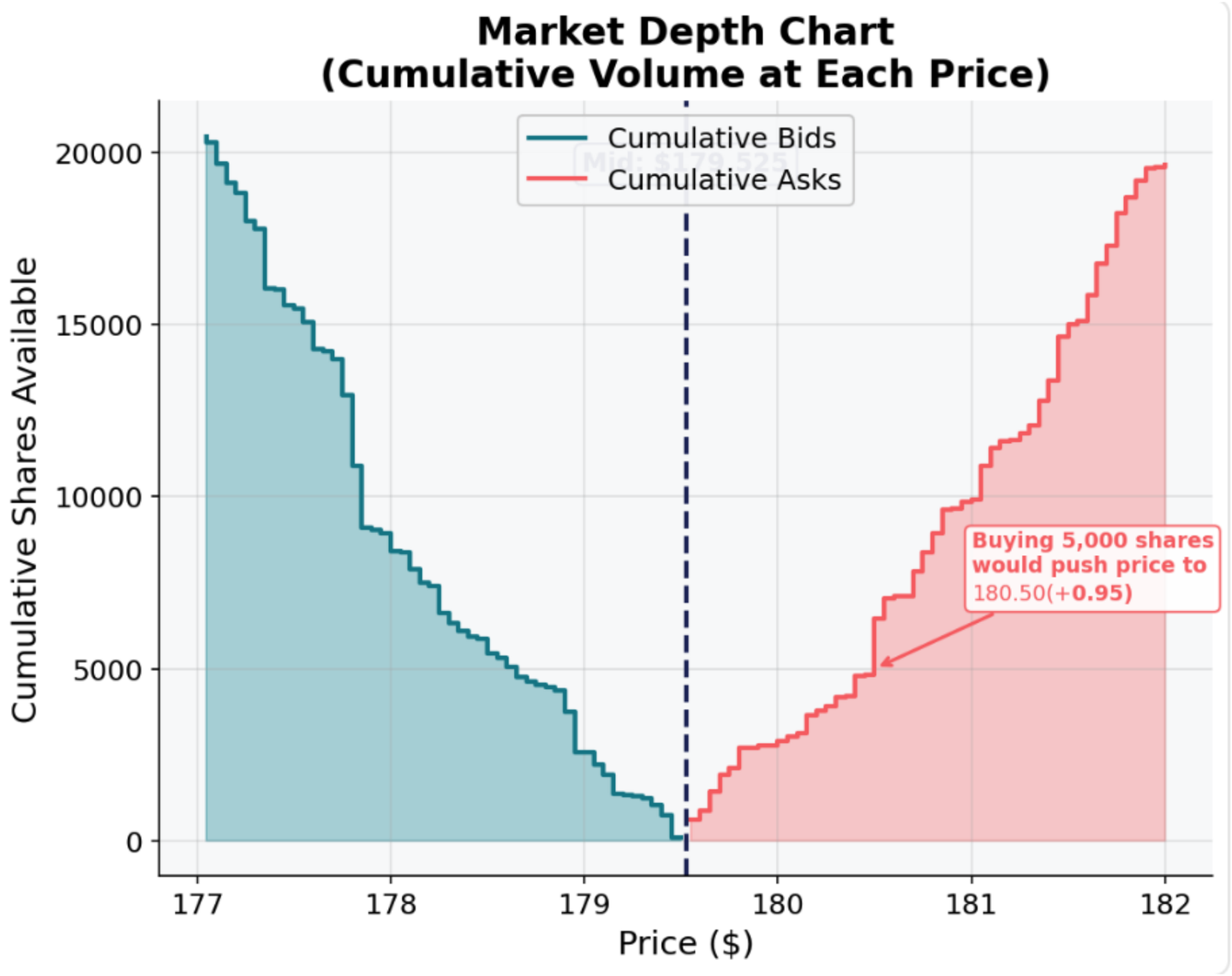
The difference between ask and bid (Ask – Bid). This spread represents an immediate cost of trading and varies by liquidity. Example: Bid $179.50, Ask $179.55, Spread $0.05

When a buyer agrees to pay the ask price or a seller accepts the bid, a transaction occurs and "the price" updates. This process happens millions of times daily in liquid markets.

Bid-Ask Price Mechanics



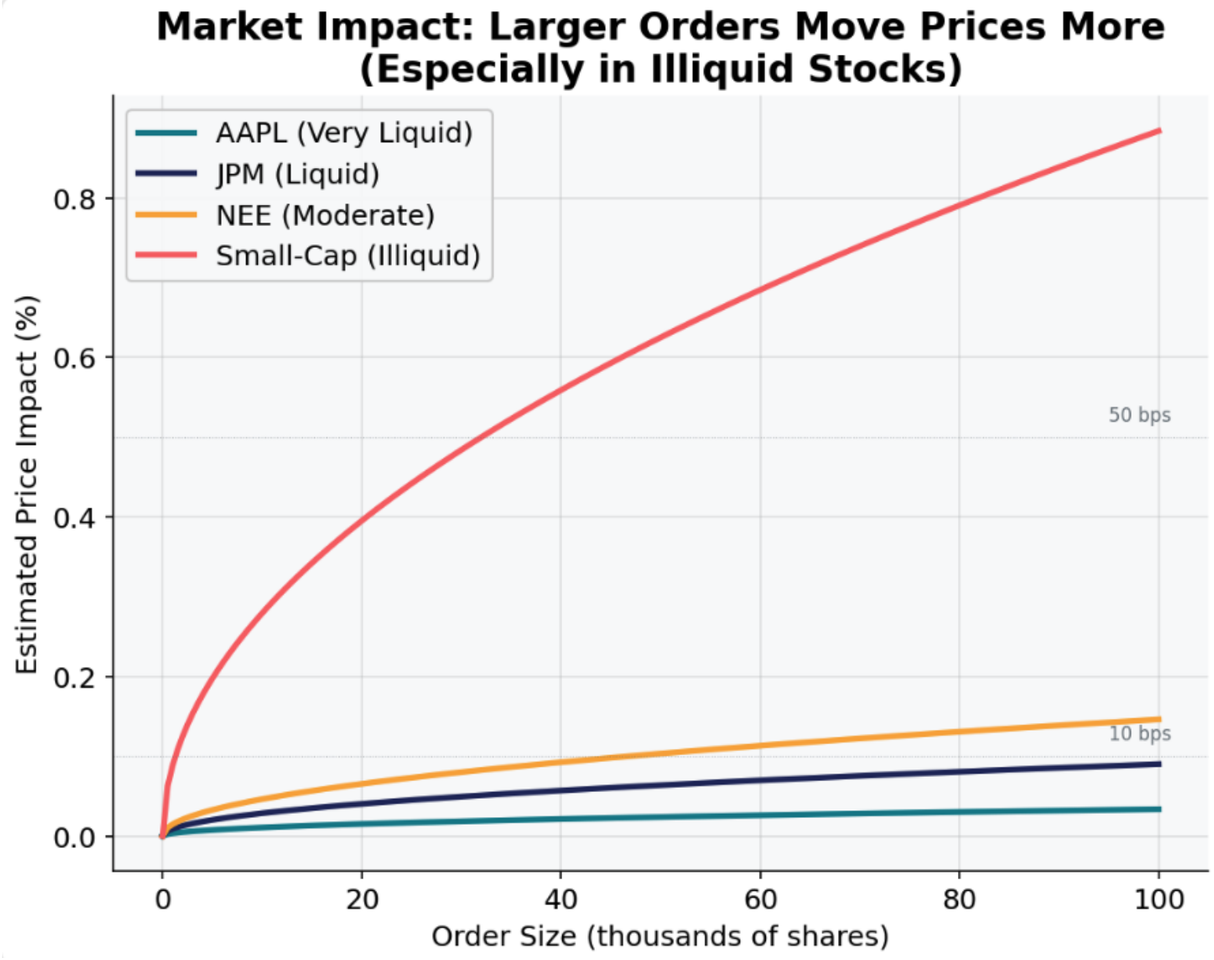
Bid-Ask Price Mechanics



The Marginal Trader Principle

Stock prices are not determined by the average opinion of all investors. Instead, they're set by the **marginal buyer** (willing to pay the most right now) and the **marginal seller** (willing to accept the least right now).

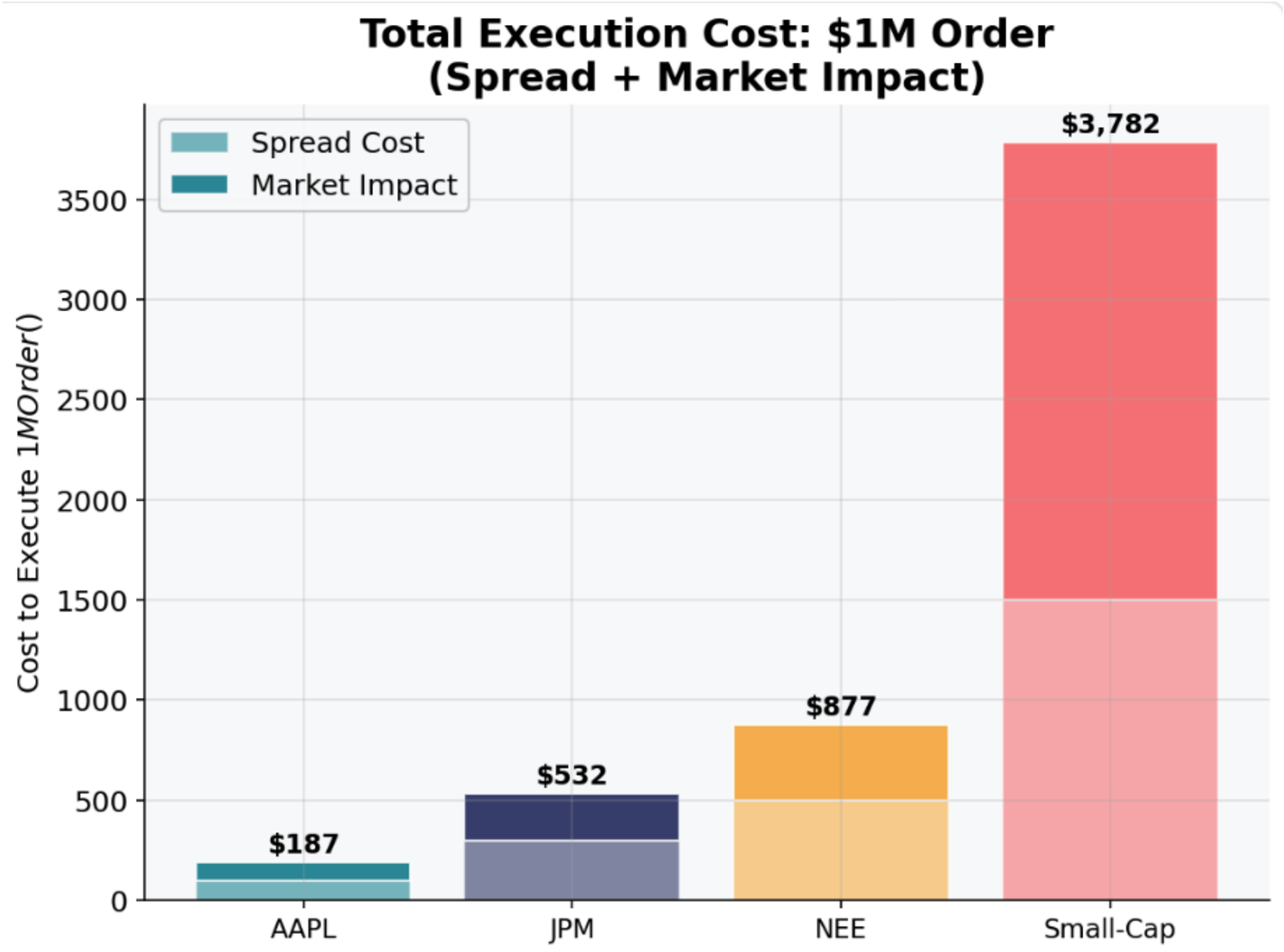
Apple doesn't need millions of investors to believe it's worth $180 per share. It only needs one marginal buyer willing to pay $180 and one marginal seller willing to sell at $180. That single transaction becomes "the price."



The Marginal Trader Principle

Sharp price movements can occur on low trading volume when marginal traders shift

AI models must predict marginal behavior, not average sentiment



Understanding the Order Book

The order book is the real-time ledger of all outstanding buy and sell orders at different price levels. It reveals market depth and liquidity—critical information for algorithmic traders.

Sample Order Book Structure

500 shares @ $179.50	300 shares @ $179.55
1,000 shares @ $179.45	600 shares @ $179.60
2,000 shares @ $179.40	1,500 shares @ $179.65

Order Book Dynamics

Depth: The quantity of shares available at each price level. Greater depth means more shares ready to trade.

Liquidity: Deep order books with narrow spreads indicate high liquidity—large orders can execute without significant price impact.

Market Impact: Thin order books mean small orders can cause disproportionate price movements, increasing execution costs.

AI Application: High-frequency trading firms employ machine learning to model order book dynamics and predict short-term price movements. These models analyze order flow patterns, cancellation rates, and book imbalances to gain microsecond advantages.

Market Microstructure for AI Practitioners

Understanding how prices form at the microstructure level is not merely academic—it's essential for developing profitable AI trading strategies. Even the most sophisticated predictive model will fail if it ignores execution realities.

Transaction Costs

The bid-ask spread represents a real cost that erodes trading profits. For high-frequency strategies, this spread can exceed expected returns, rendering otherwise profitable signals useless.

Market Impact

Large orders move prices against you as you execute—buying pushes prices up, selling pushes them down. Optimal execution algorithms must minimize this impact.

Liquidity Variation

Trading Apple (millions of daily shares) differs fundamentally from trading small-cap stocks (thousands of shares). AI strategies must adapt to liquidity constraints.

Price Discovery

Markets aggregate information from millions of participants globally. Understanding this process helps AI models interpret price signals and distinguish noise from information.

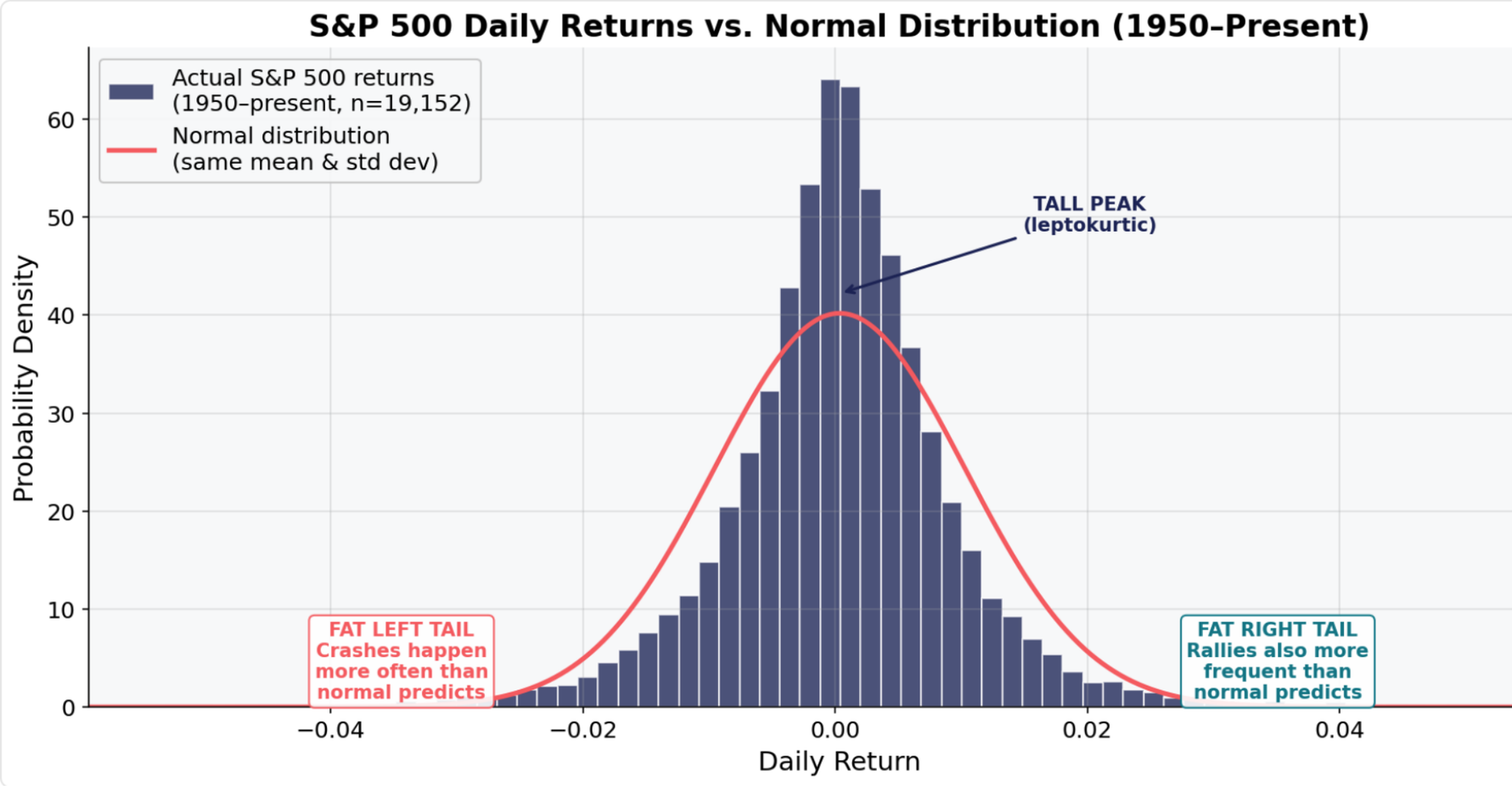
The Normal Distribution Assumption

Theoretical Expectations

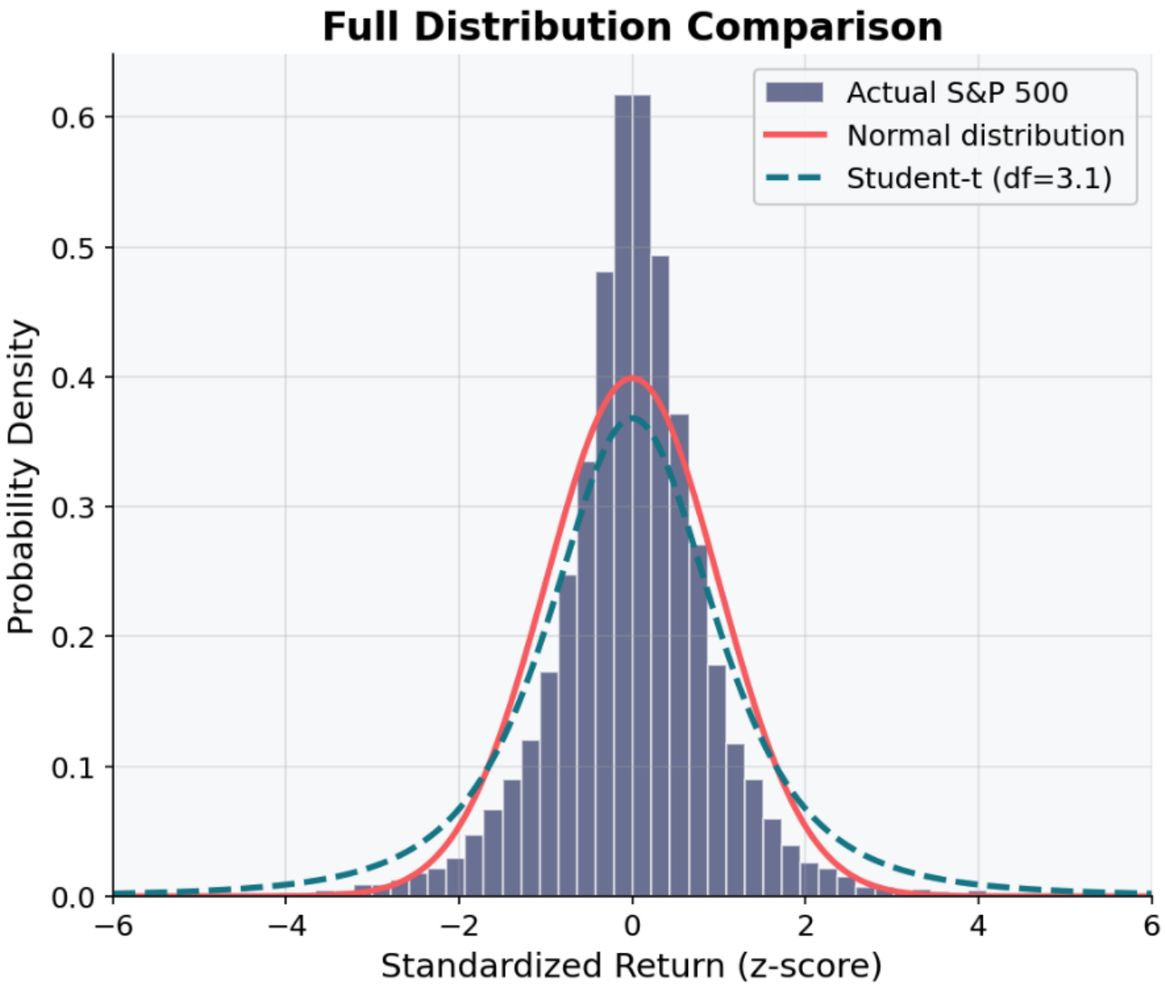
The **normal distribution** (Gaussian bell curve) underpins most classical finance models:

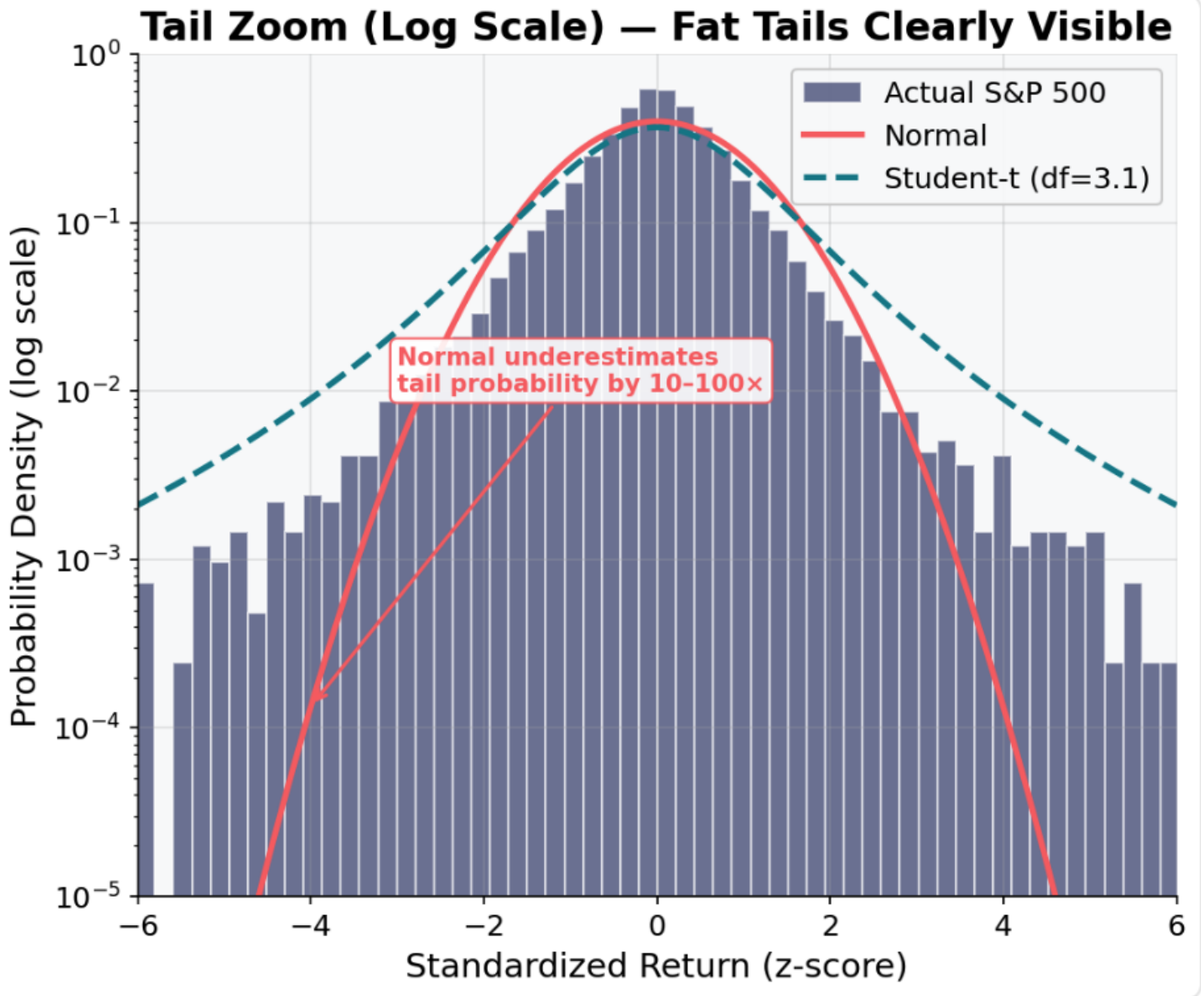
- Symmetric around the mean
- 68% of observations within 1 standard deviation
- 95% within 2 standard deviations
- 99.7% within 3 standard deviations

Extreme events beyond 3 standard deviations should be extraordinarily rare under this assumption.



Fat Tails of Returns





The Reality: Fat Tails

Real financial markets exhibit **fat tails** (heavy tails)—extreme events occur far more frequently than the normal distribution predicts. This phenomenon has repeatedly caught financial institutions and quantitative models off guard.

The Frequency Problem

A "6-sigma event" should occur once every 4 million days under normal distribution—yet financial markets experience such moves every few years.

The 2008 Crisis

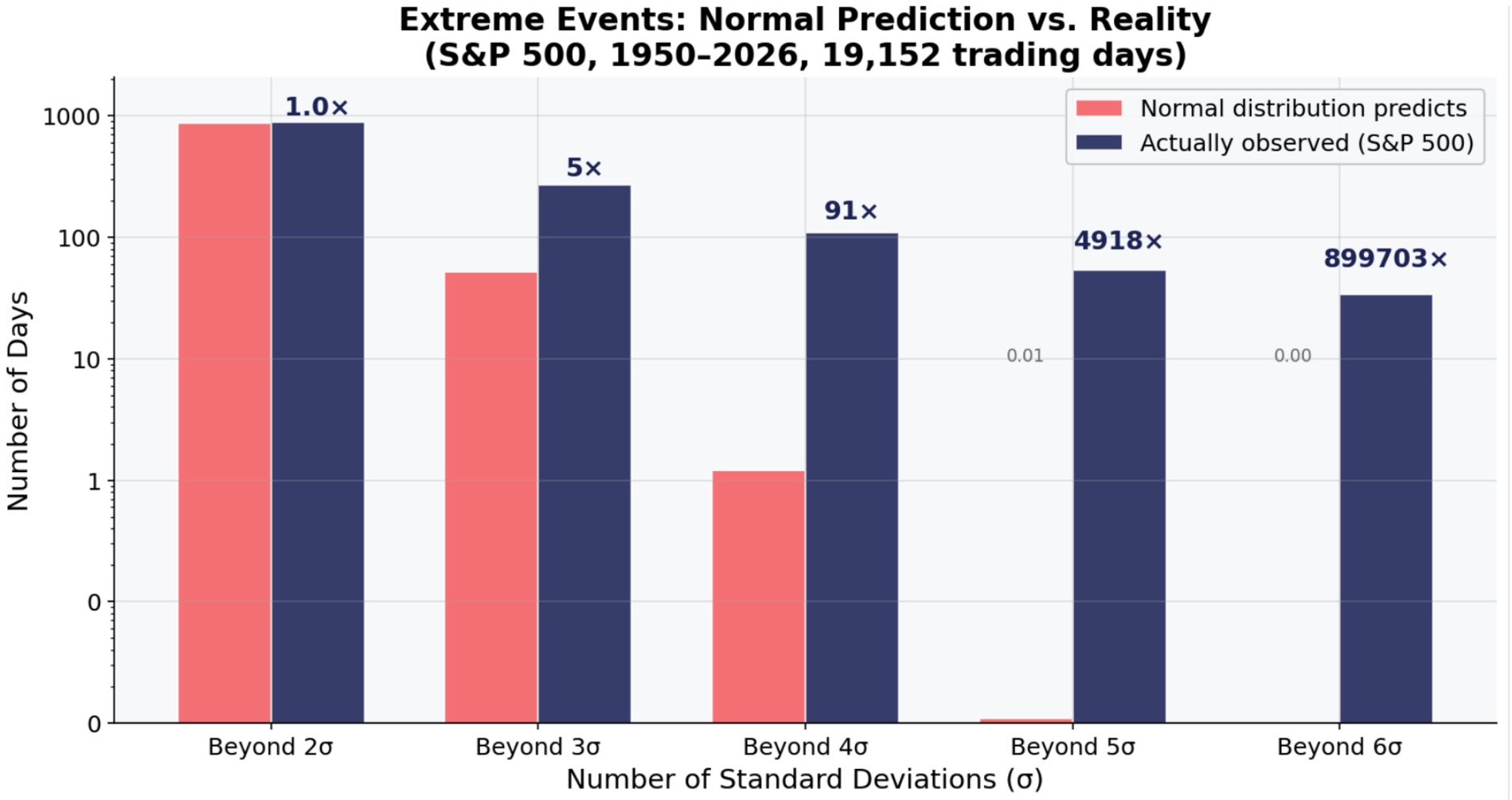
The financial crisis registered as a 25-sigma event according to normal distribution assumptions—a probability so small it's essentially impossible, yet it happened.

Leptokurtic Character

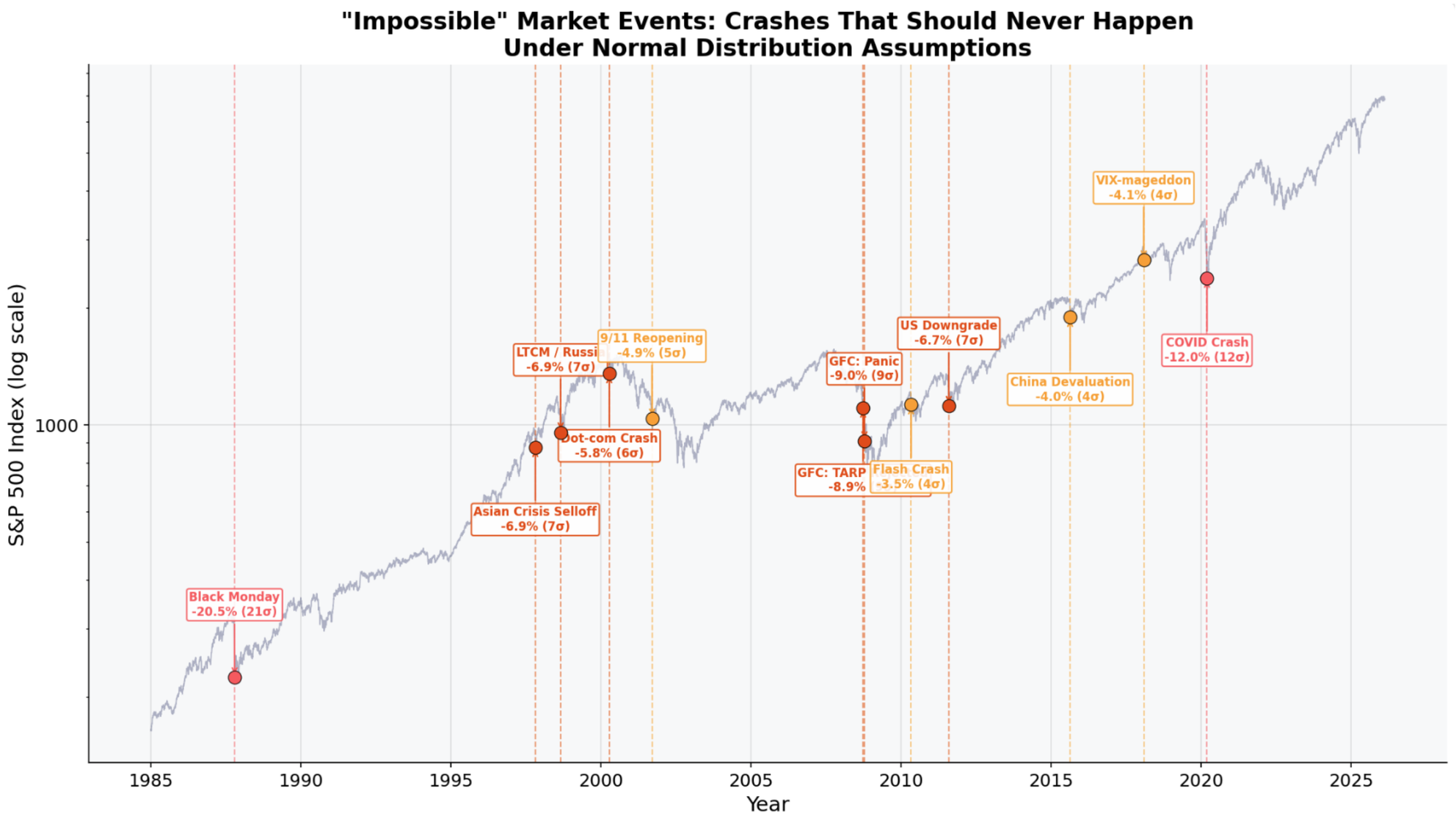
Actual return distributions show taller peaks (more "normal" days) and fatter tails (more extreme days)—exhibiting positive excess kurtosis.

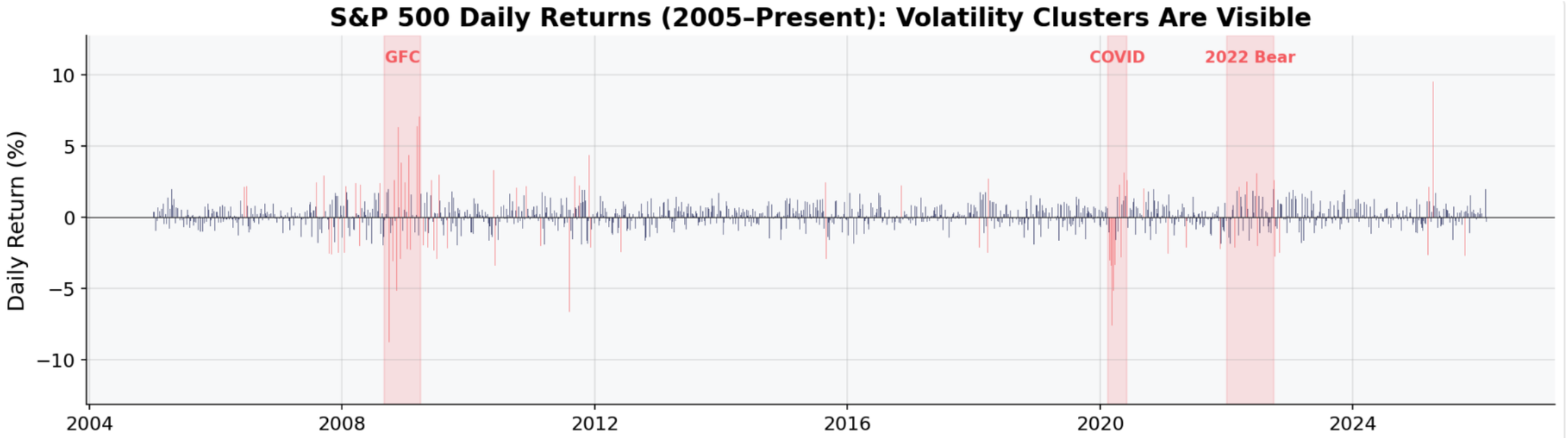
This systematic underestimation of tail risk represents one of the greatest challenges—and opportunities—for AI in finance. Machine learning models that properly account for fat tails can provide more realistic risk assessments than traditional approaches.

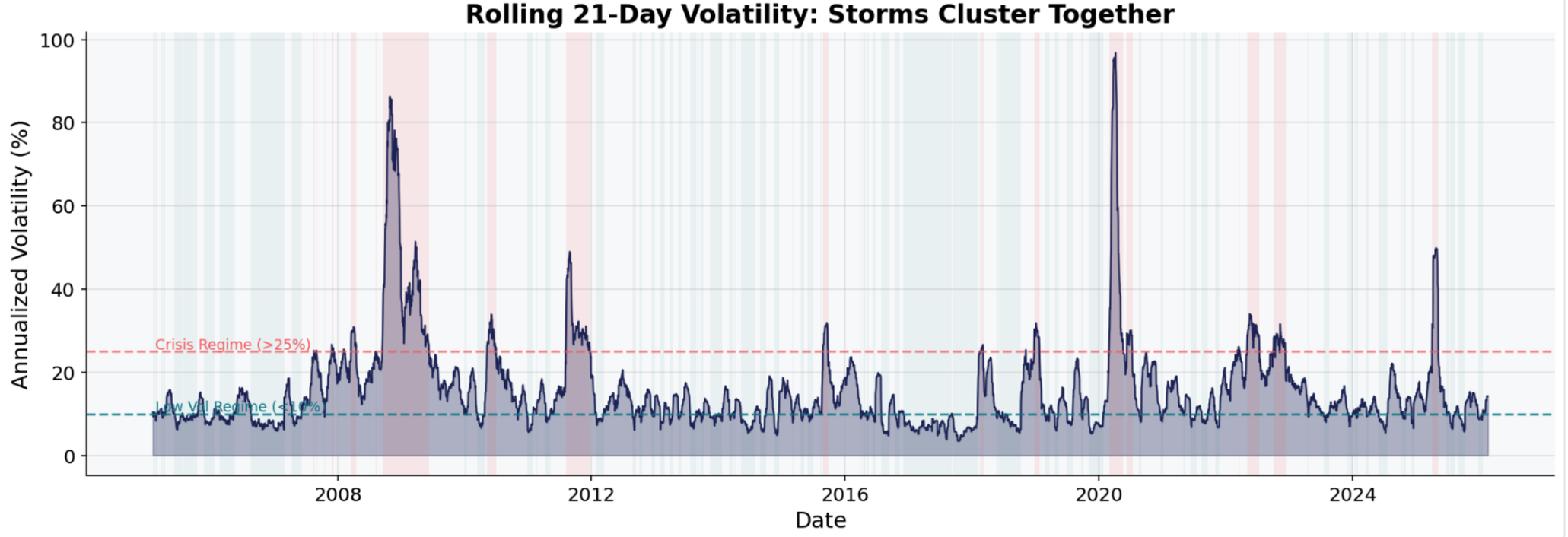
Expect the Unexpected



Impossible Events Occur Regularly




S&P 500 Daily Returns (2005–Present): Volatility Clusters Are Visible
GFC
COVID
2022 Bear
Daily Return (%)
10
5
0
−5
−10
2004
2008
2012
2016
2020
2024


Rolling 21-Day Volatility: Storms Cluster Together
Annualized Volatility (%)
100
80
60
40
20
0
Crisis Regime (>25%)
Low Vol Regime (<10%)
2008
2012
2016
2020
2024
Date

Normal Returns vs. Log-Normal Prices

Understanding the relationship between returns and prices requires distinguishing between two related but distinct statistical concepts—a distinction with important implications for how we model financial data.

Returns: Approximately Normal

Simple returns measure percentage changes and can be positive or negative, centered around a small positive mean, and roughly symmetric (though with those problematic fat tails we discussed).

Formula for simple return:

$$\frac{P_{today} - P_{yesterday}}{P_{yesterday}}$$

Prices: Log-Normal Distribution

Price levels follow a different pattern—they can never fall below zero (you can't lose more than 100% of your investment), can double or triple upward, and exhibit right-skewed distributions.

Formula for log return:

$$\ln\left(\frac{P_{today}}{P_{yesterday}}\right)$$

Why Log Returns Matter

For small changes, log returns approximate simple returns. But for longer horizons, log returns offer mathematical convenience—they're additive across time periods (weekly log return = sum of daily log returns) and more suitable for compounding calculations.

Implications for Risk Management

If Returns Were Normal

- Risk management would be straightforward—just multiply standard deviations
- Market crashes would be virtually impossible
- Value-at-Risk models would accurately predict losses
- Portfolio insurance costs would be minimal

The Reality of Fat Tails

Because actual returns exhibit fat tails, traditional models systematically underestimate risk. "Once in a century" events occur every decade.

Critical Takeaway: AI models must explicitly account for tail risk. This means testing distribution assumptions, implementing robust backtesting across crisis periods, and incorporating extreme value theory or other heavy-tail modeling approaches.

The Capital Asset Pricing Model

The CAPM represents one of finance's most influential frameworks, addressing a fundamental question: what return should an investor expect from a risky investment.

The CAPM Formula

$$E(R_i) = R_f + \beta_i \times (E(R_m) - R_f)$$

Where *E(R_i)* represents expected return of stock i, *R_f* is the risk-free rate (currently ~5% for Treasury bills), |*beta_i* measures the stock's market sensitivity, and ***E(R_m) - R_f*** **is the market risk premium** (historically 5-7%).

The Core Insight

Your expected return equals the risk-free rate plus compensation for systematic market risk you're bearing. The model elegantly separates risk into two components: diversifiable (firm-specific) and systematic (market-wide).

CAPM implies that only systematic risk—measured by beta—deserves compensation. Firm-specific risk can be diversified away and therefore doesn't command higher expected returns.

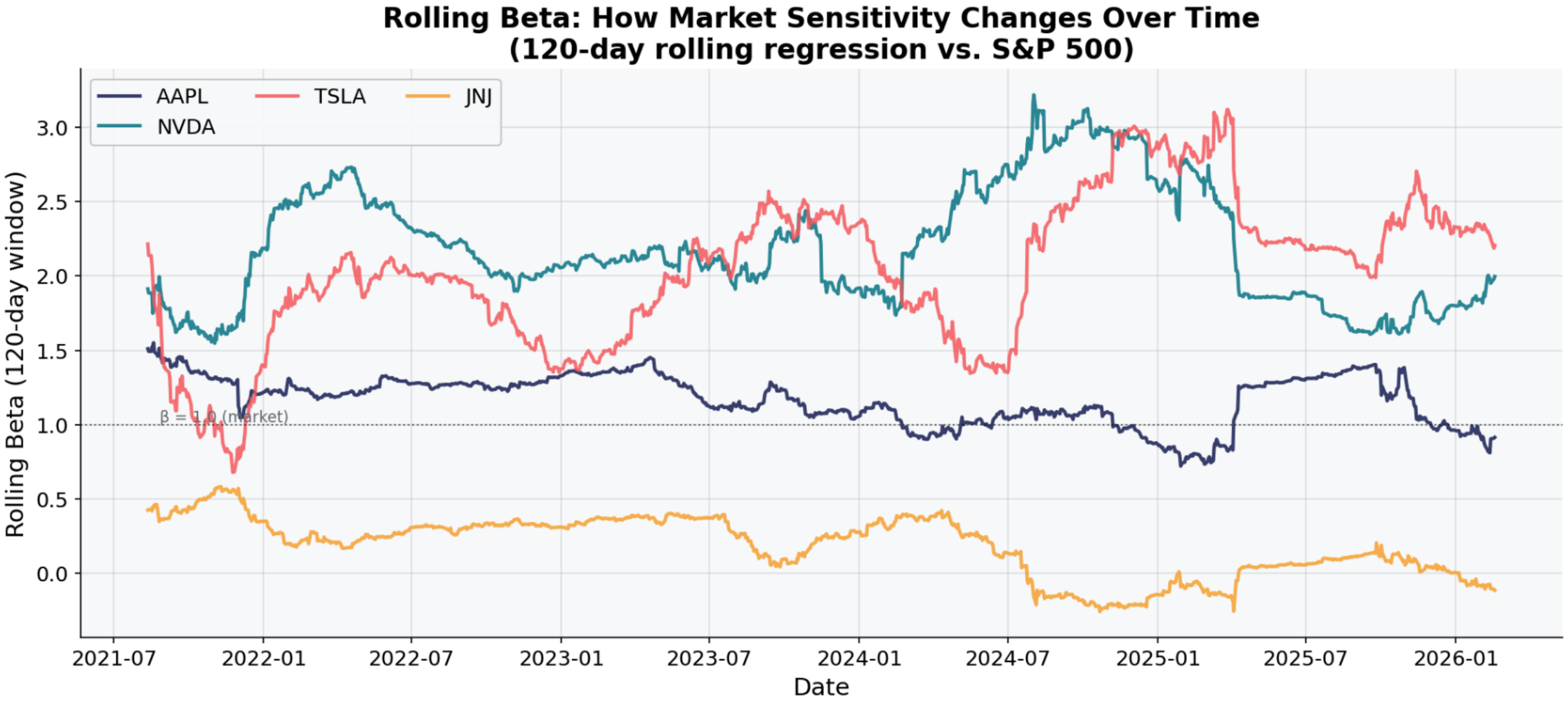
Understanding Beta

Beta quantifies how much a stock moves relative to the overall market (typically the S&P 500). It's the cornerstone metric for assessing systematic risk and forms the foundation for expected return calculations.

$\beta = 1.0$	Moves exactly with market	Index funds (SPY)
$\beta = 1.5$	50% more volatile than market	Growth tech (NVDA)
$\beta = 0.5$	50% less volatile than market	Utilities (NEE)
$\beta = 0$	Uncorrelated with market	Market-neutral strategies
$\beta < 0$	Moves opposite to market	Gold, VIX (sometimes)

Calculating Beta: Run a regression of stock returns against market returns. Beta equals the slope of the regression line. In today's lab, we'll compute beta for real stocks and interpret the results.

Rolling Beta Over Time for Selected Stocks



Alpha: The Holy Grail

Alpha represents the return above or below what CAPM predicts—the difference between actual performance and expected performance given the stock's beta. It's what separates skill from mere risk-taking.

1

Positive Alpha Example

Stock with β=1.2 should return 11% per CAPM, but actually returns 14%

Alpha = 14% − 11% = **+3%**

2

Negative Alpha Example

Same stock returns only 8%

Alpha = 8% − 11% = **−3%**

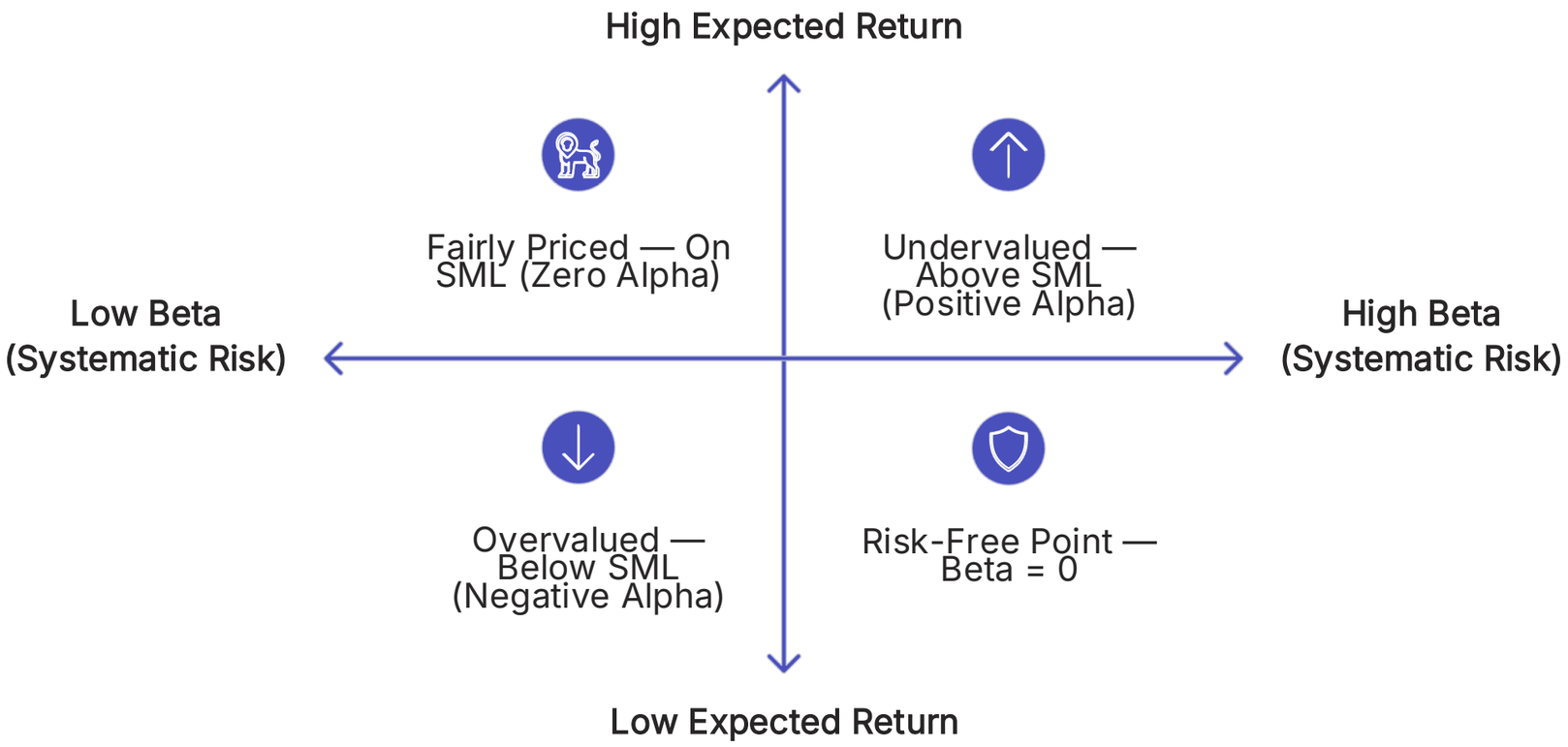
The Alpha Challenge

Alpha represents genuine skill (or luck) beyond simply taking market risk. The entire active investment industry—hedge funds, stock pickers, tactical allocation—exists to generate positive alpha.

Most active managers fail to generate positive alpha consistently after fees. Studies show only 5-10% of actively managed funds outperform their benchmarks over 10+ year periods.

The AI Promise: Can machine learning identify persistent alpha that human analysis misses?

The Security Market Line

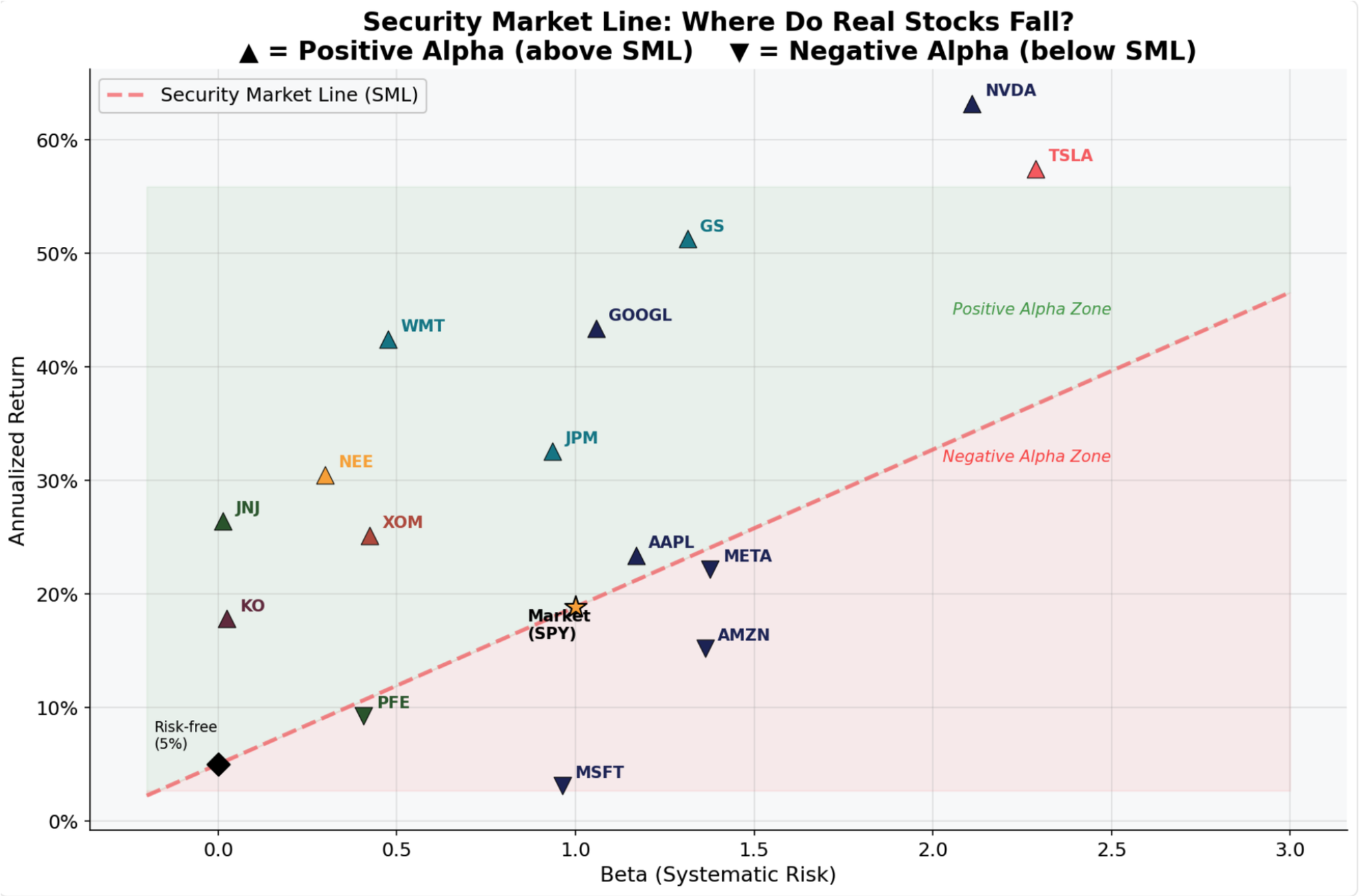


The SML is a straight line starting from the risk-free rate and rising with beta. Its slope represents the market risk premium—the compensation investors demand for bearing systematic risk.

On the Line
Fairly priced securities where actual return matches CAPM prediction

Above the Line
Undervalued securities with positive alpha—generating excess return relative to risk

Below the Line
Overvalued securities with negative alpha—insufficient return for the risk taken


Security Market Line: Where Do Real Stocks Fall?
▲ = Positive Alpha (above SML) ▼ = Negative Alpha (below SML)
Security Market Line (SML)
Positive Alpha Zone
Negative Alpha Zone
NVDA
TSLA
GS
GOOGL
WMT
JPM
NEE
JNJ
XOM
AAPL
META
KO
Market (SPY)
AMZN
PFE
Risk-free (5%)
MSFT
Annualized Return
Beta (Systematic Risk)
0%
10%
20%
30%
40%
50%
60%
0.0
0.5
1.0
1.5
2.0
2.5
3.0

Introduction to Options

Options are derivative contracts that provide asymmetric risk-reward profiles—they give holders rights without obligations, creating powerful portfolio management and speculation tools.

Option Types

Call Option

Right to **buy** the underlying asset at the strike price. Benefits from price increases.

Put Option

Right to **sell** the underlying asset at the strike price. Benefits from price decreases.

Key Terminology

- **Strike Price (K):** The fixed price specified in the contract
- **Expiration Date:** When the option contract expires
- **Premium:** Price paid to purchase the option
- **Underlying:** The stock or asset the option is based on
- **Exercise:** Using your right to buy or sell

Think of a put option as portfolio insurance—you pay a premium to protect against downside. A call option is like a house deposit—you lock in a purchase price today with limited upfront cost.

Call Option Mechanics

Let's walk through a concrete example to understand how call options work and when they become profitable. This will illustrate the asymmetric payoff structure that makes options powerful.

Example Trade Setup

You purchase a call option on Apple stock with these terms: **Strike Price:** $180 | **Premium Paid:** $5 per share | **Expiration:** 1 month from today

Scenario 1: Strong Rally

Apple rises to $200. You exercise: buy at $180, immediately worth $200. **Profit:** $200 – $180 – $5 premium = **$15 per share**

Scenario 2: Modest Gain

Apple rises to $190. You exercise: buy at $180, worth $190. **Profit:** $190 – $180 – $5 premium = **$5 per share**

Scenario 3: Price Decline

Apple falls to $175. No point exercising—you can buy cheaper in the open market. You simply let the option expire. **Loss:** $5 per share (just the premium paid)

The Key Insight: Your downside is strictly limited to the premium paid ($5), but your upside is theoretically unlimited. This asymmetric payoff—capped losses with unlimited gains—makes options attractive for speculation and hedging.

Put Option Mechanics

Put options work inversely to calls, providing profit when the underlying asset declines. They're essential for portfolio hedging and speculation on downside moves.

Example Trade Setup

You purchase a put option on Apple stock: **Strike Price:** $180 | **Premium Paid:** $4 per share | **Expiration:** 1 month from today

Scenario 1: Price Decline

Apple drops to $160. You exercise your right to sell at $180, even though market price is only $160.

Profit: $180 – $160 – $4 premium = **$16 per share**

This demonstrates how puts provide insurance against downside risk—the further prices fall, the more valuable your put becomes.

Scenario 2: Price Increase

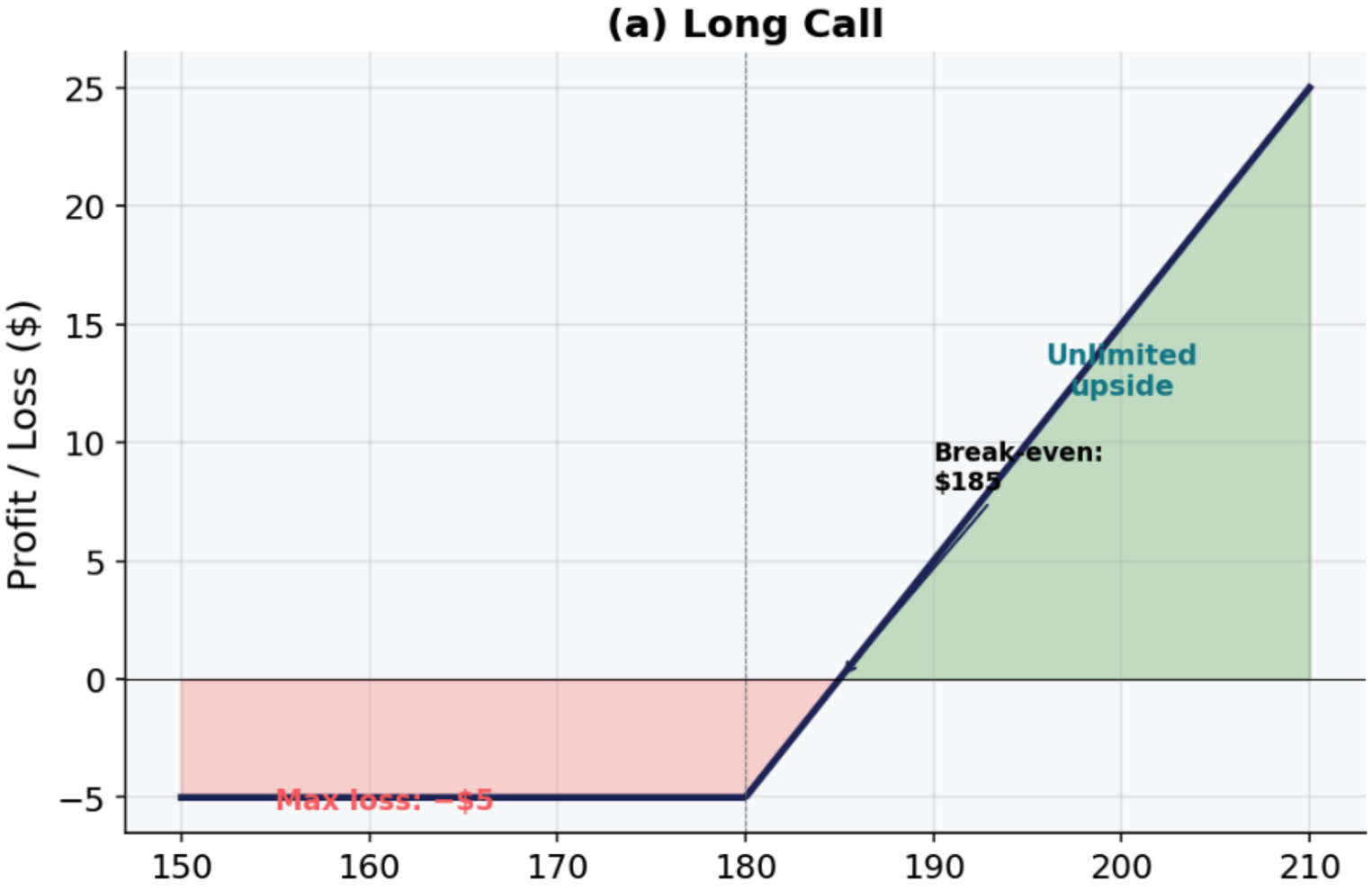
Apple rises to $185. No point exercising—you can sell for more in the open market. You let the option expire worthless.

Loss: $4 per share (just the premium)

Like insurance that expires unused, you lose the premium but protected yourself against potential catastrophic losses.

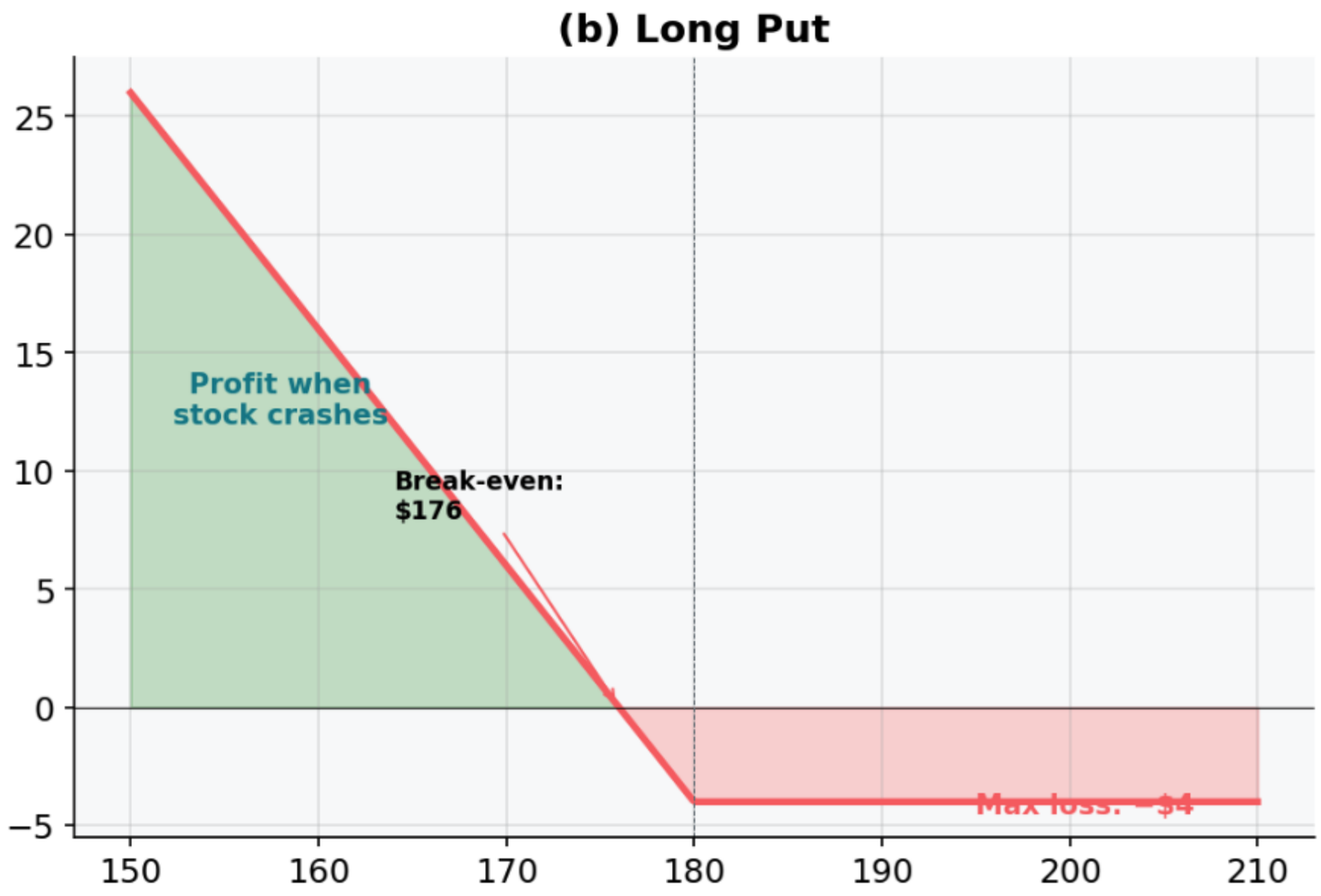
Puts function as portfolio insurance. Fund managers regularly purchase puts to protect large equity positions against market crashes, accepting the premium cost as insurance expense.

Long Call Payoff



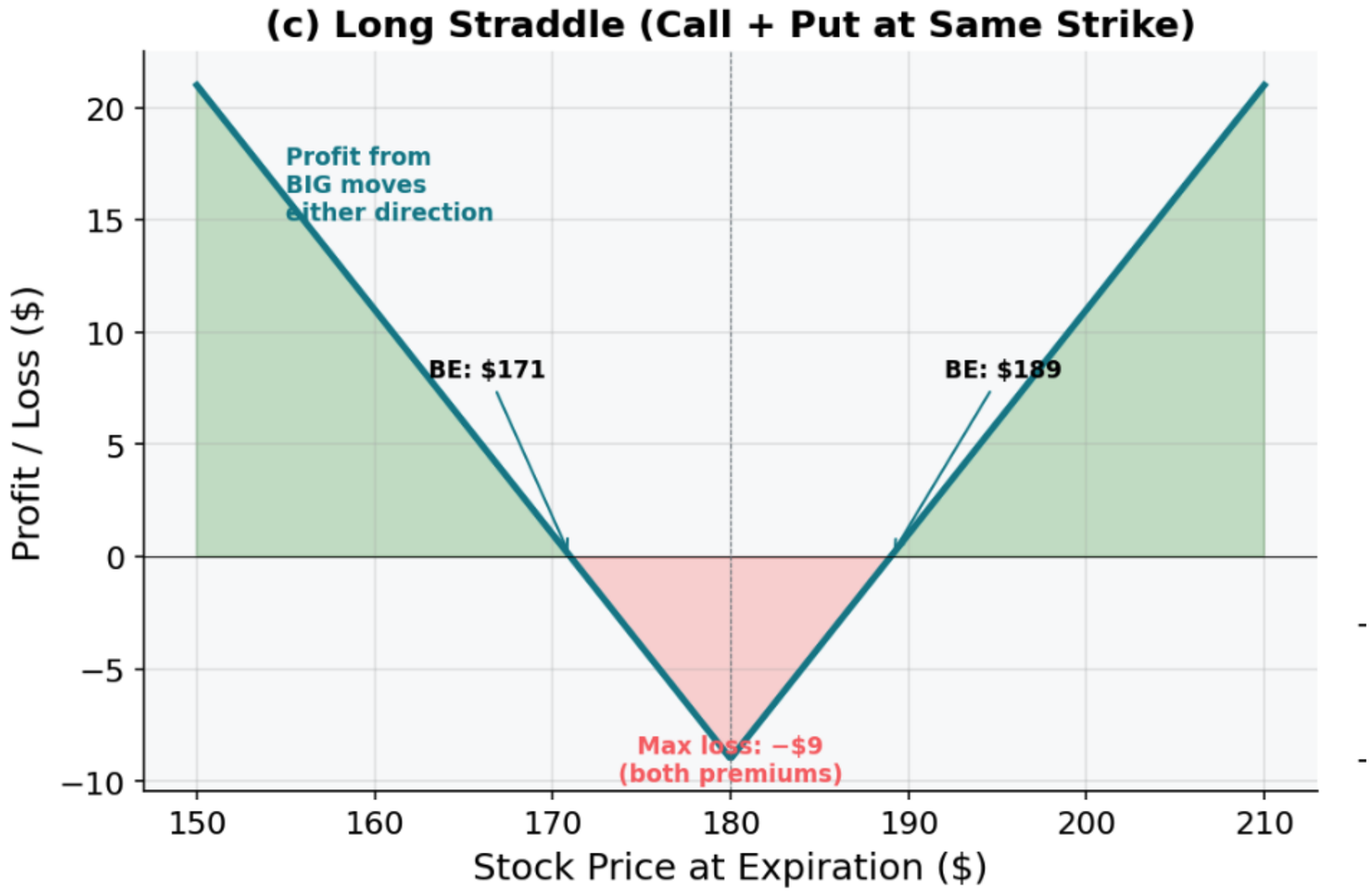
- The call option has a cost of $5 and a strike price of 180
- Any price lower than $180 – option not exercised (worthless)
- Any price above 185 option value is price - 180

Long Put Payoff

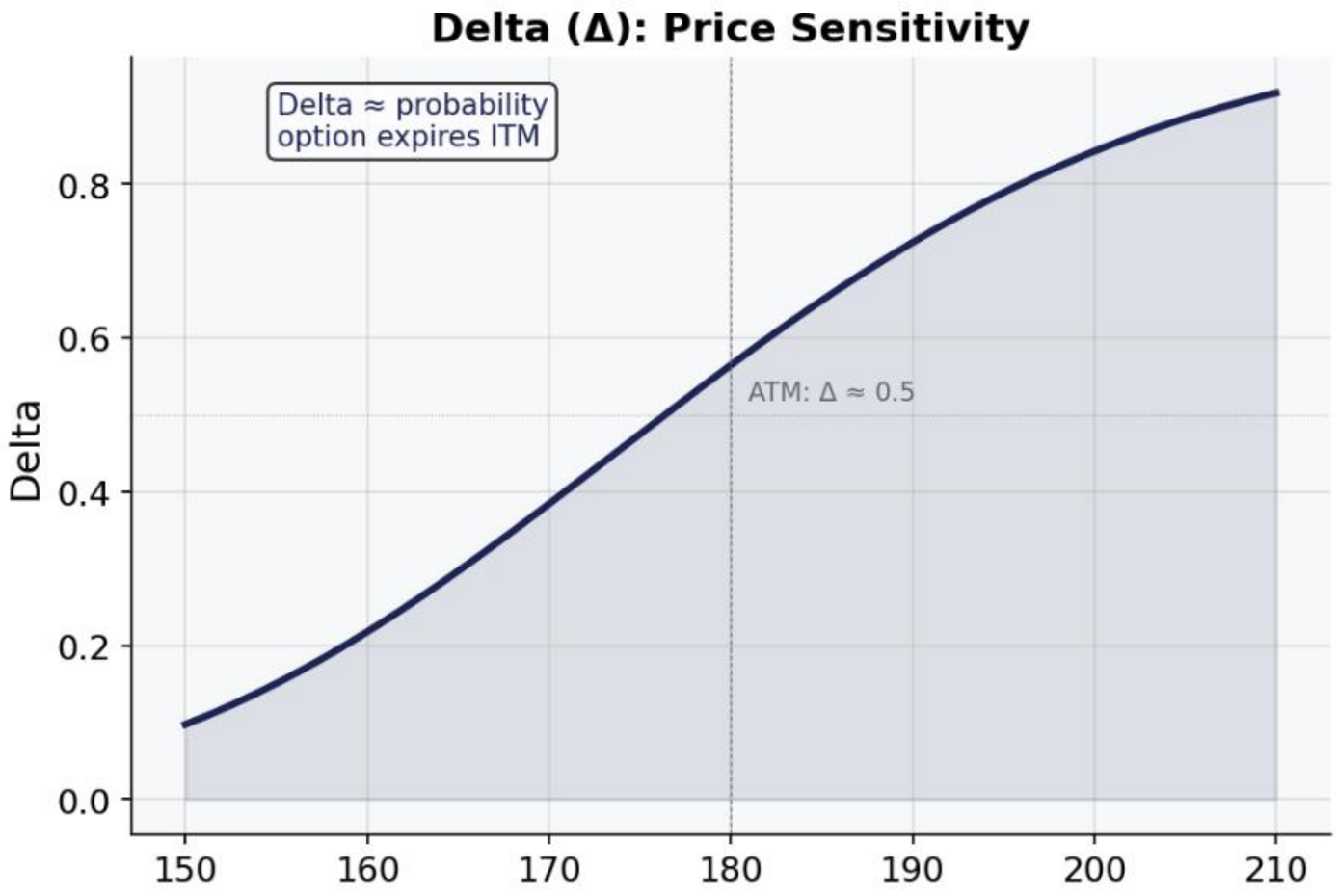


- Put is a bet the stock price will decline
- Here we purchase a $4 put with a strike price of $180
- Breakeven price of $176
- Profit when stock price declines

Many Variation in Options

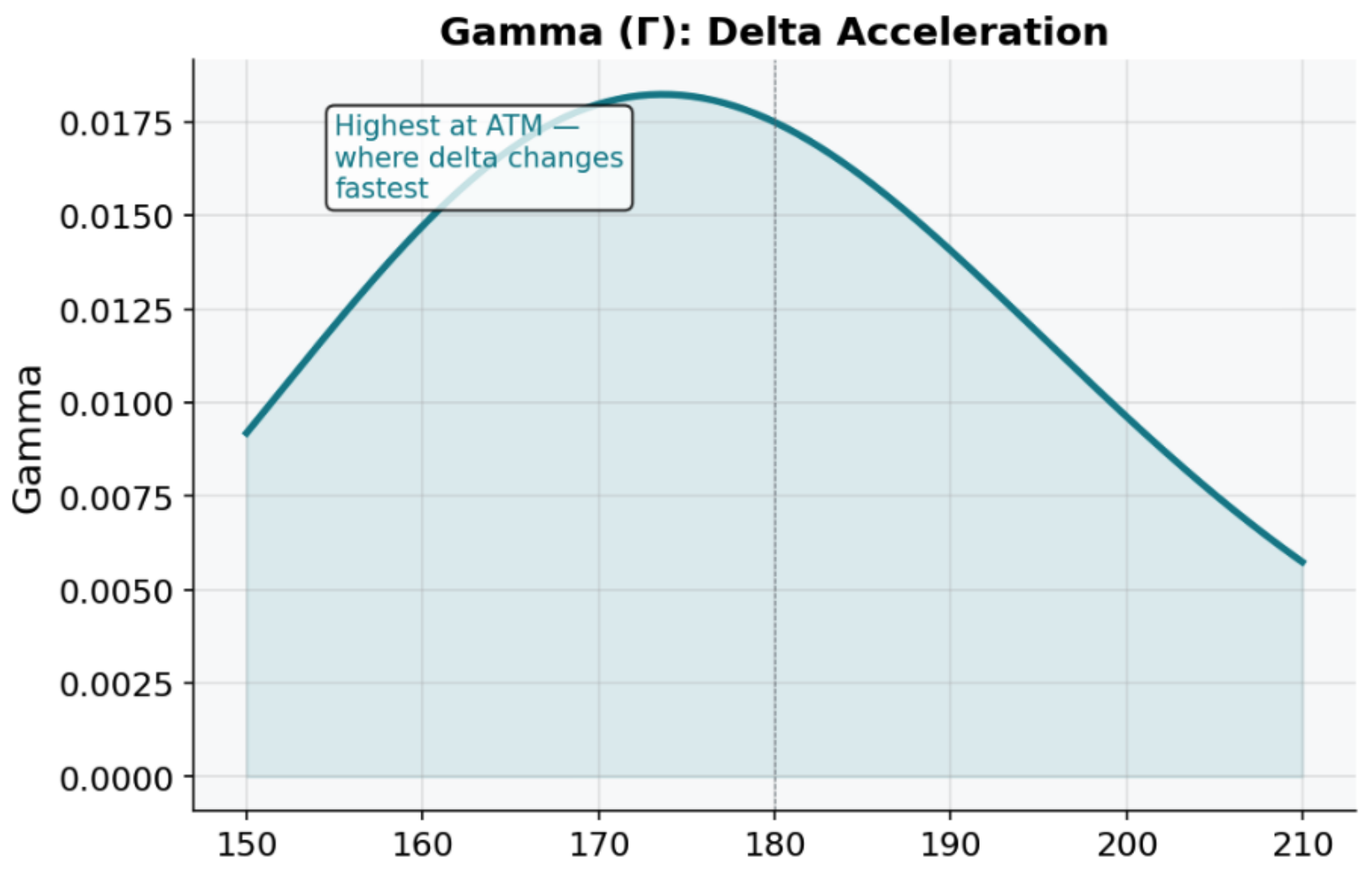


Greeks: How A Call Price Responds to Market Changes



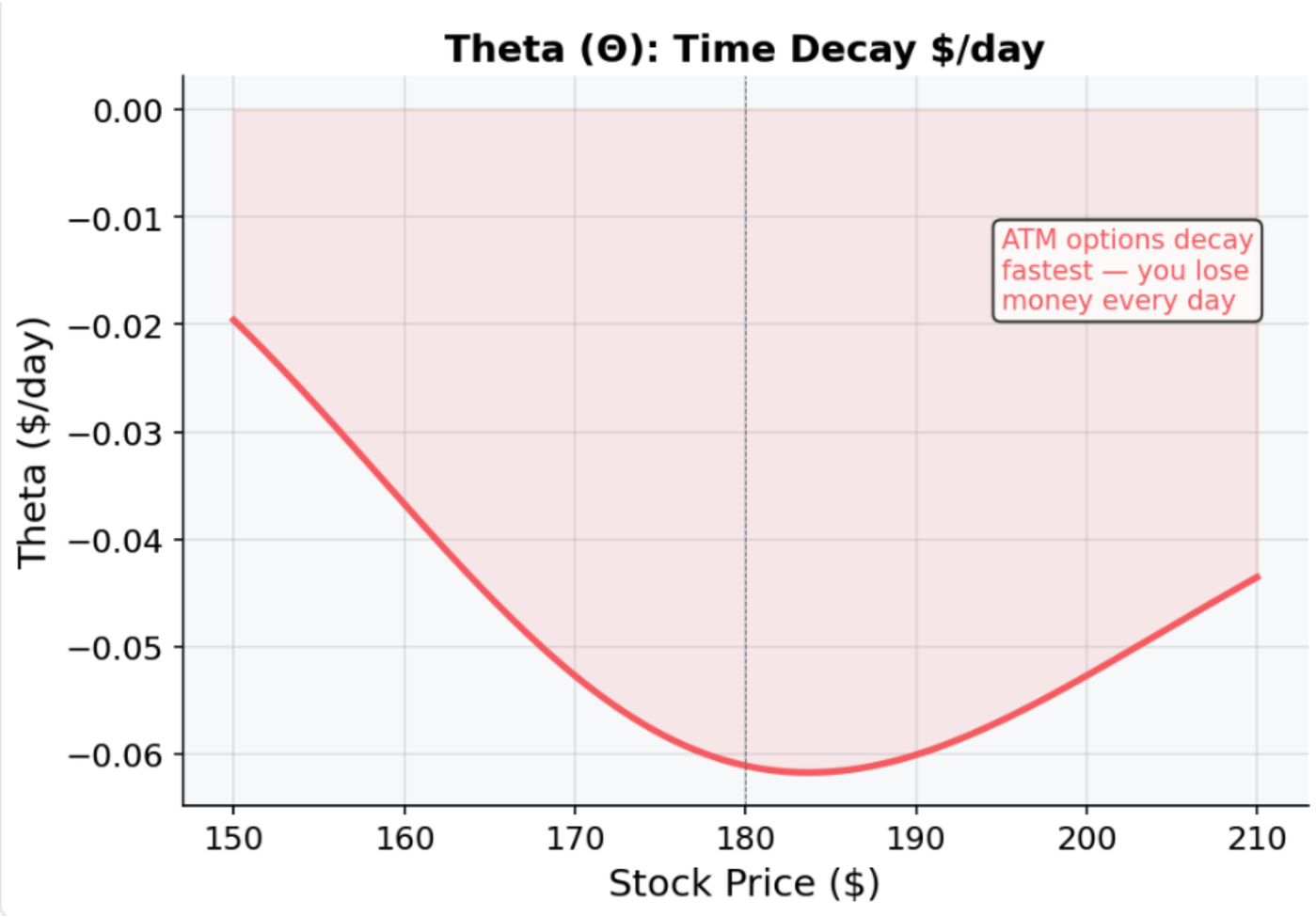
- K - $180 strike price, T = 3 months to expiration, vol of 25% and r = 5 RF rate
- How much the option price moves for a $1 move in the stock
- At-the-money (ATM) options have delta ~ 0.5
- Deep in-the-money → delta near 1 (moves like the stock); deep out-of-the-money → delta near 0
- Intuition: roughly the probability the option expires in the money

Greeks: How A Call Price Responds to Market Changes



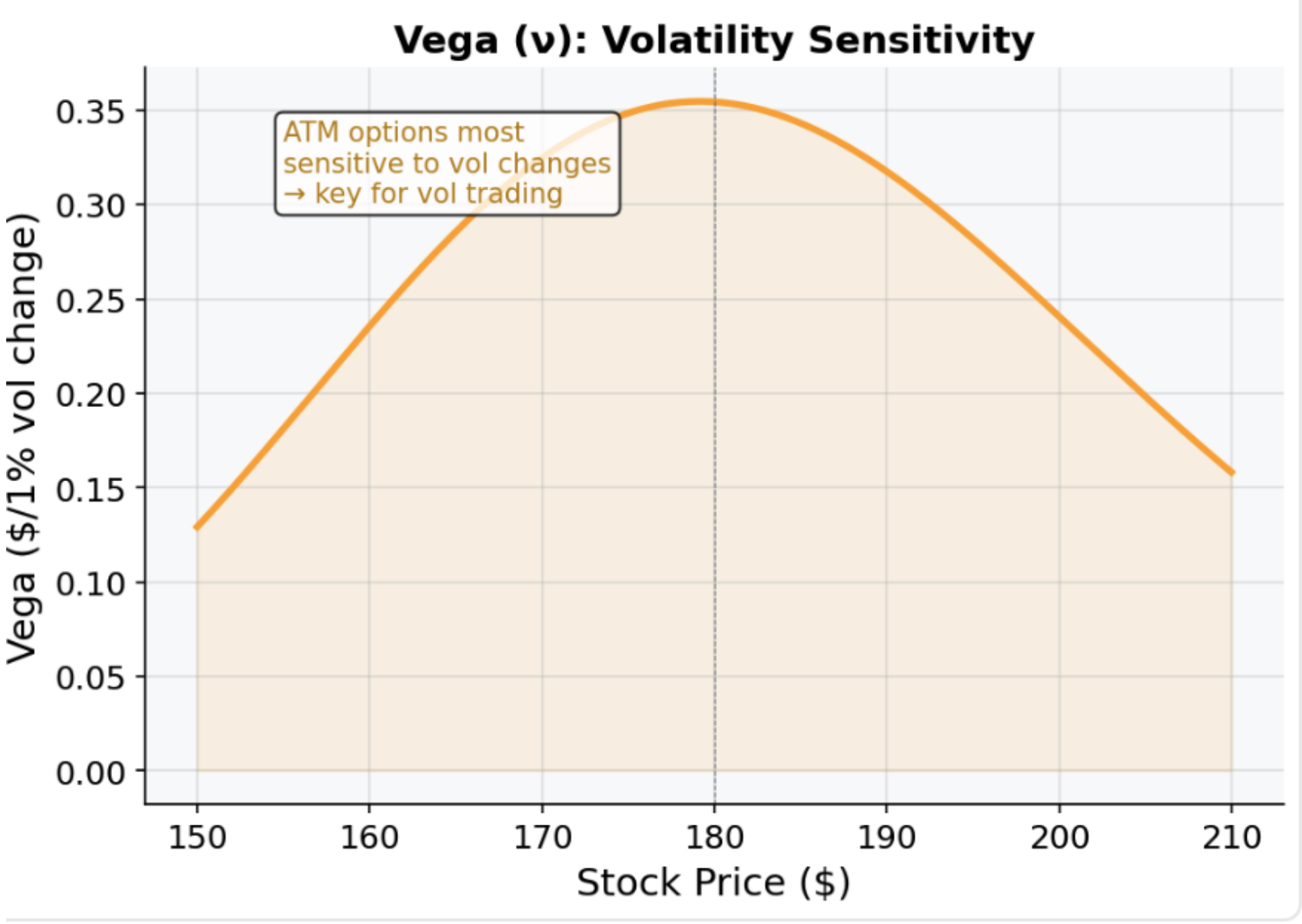
- K - $180 strike price, T = 3 months to expiration, vol of 25% and r = 5 RF rate
- How much delta itself changes as the stock price moves (the "speed" of delta)
- Highest at ATM — this is where delta is most unstable and hardest to hedge
- Important for traders who delta-hedge: high gamma means you need to rebalance frequently

Greeks: How A Call Price Responds to Market Changes



- How much value the option loses each day just from time passing
- Always negative for option buyers — you're losing money every day you hold
- Decay is fastest for ATM options and accelerates as expiration approaches
- Intuition: options are "melting ice cubes"

Greeks: How A Call Price Responds to Market Changes

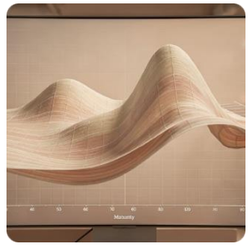


- How much the option price changes for a 1% increase in implied volatility
- Highest at ATM — those options are most sensitive to volatility swings
- Rising volatility increases option value (more chance of a big move); falling volatility hurts option holders
- Key metric for traders who trade volatility rather than direction

Options in AI Finance

Enhanced Pricing Models

Black-Scholes assumes normal returns, but we've seen returns have fat tails. Machine learning can incorporate skewness, other distributional features for more accurate pricing.



Implied Volatility Analysis

Option prices embed the market's expectation of future volatility. AI models can extract signals from implied volatility patterns, term structures, and cross-sectional relationships.

Dynamic Hedging

Portfolio insurance using options requires constant recalculation as prices and volatility change. This continuous optimization problem is ideal for algorithmic approaches.

Strategy Generation

Complex option strategies—spreads, butterflies, condors, straddles—offer millions of possible combinations.

Portfolio Diversification Preview

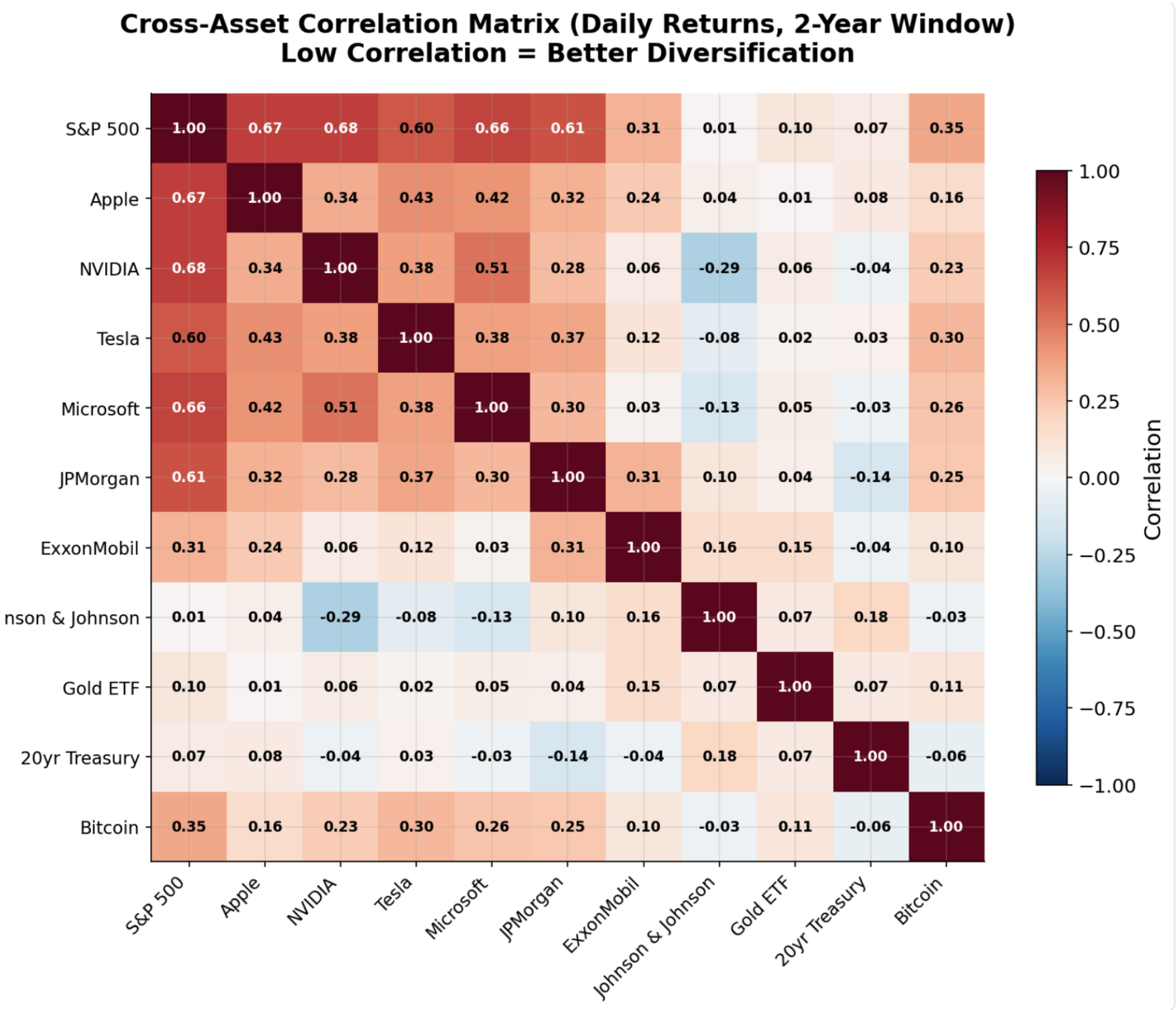
Harry Markowitz revolutionized finance in 1952 with a deceptively simple insight: combining assets that aren't perfectly correlated reduces portfolio risk below the weighted average of individual asset risks. This phenomenon—diversification—is often called the only "free lunch" in finance.

If you hold just Apple stock, your risk equals Apple's volatility. But if you hold Apple plus Microsoft, and they don't always move together, your combined risk is less than the average of the two. This mathematical property becomes more powerful as you add assets with low correlations.

Mean-Variance Framework

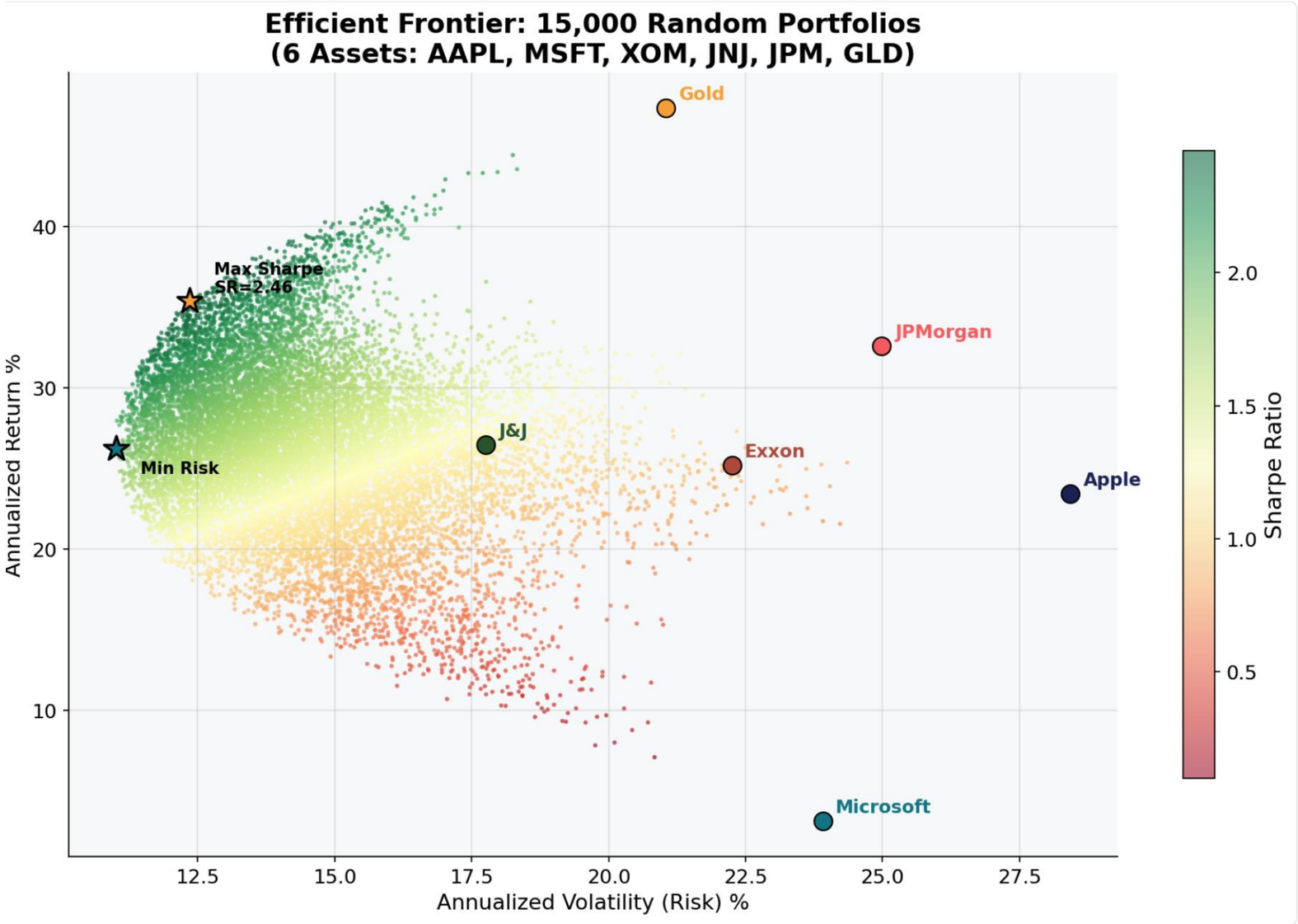
Markowitz formalized this intuition mathematically. For any target return level, there exists an optimal combination of assets that minimizes portfolio volatility. The set of these optimal portfolios forms the **efficient frontier**—portfolios offering the highest return for a given risk level.

This breakthrough earned Markowitz the Nobel Prize and spawned modern portfolio theory.



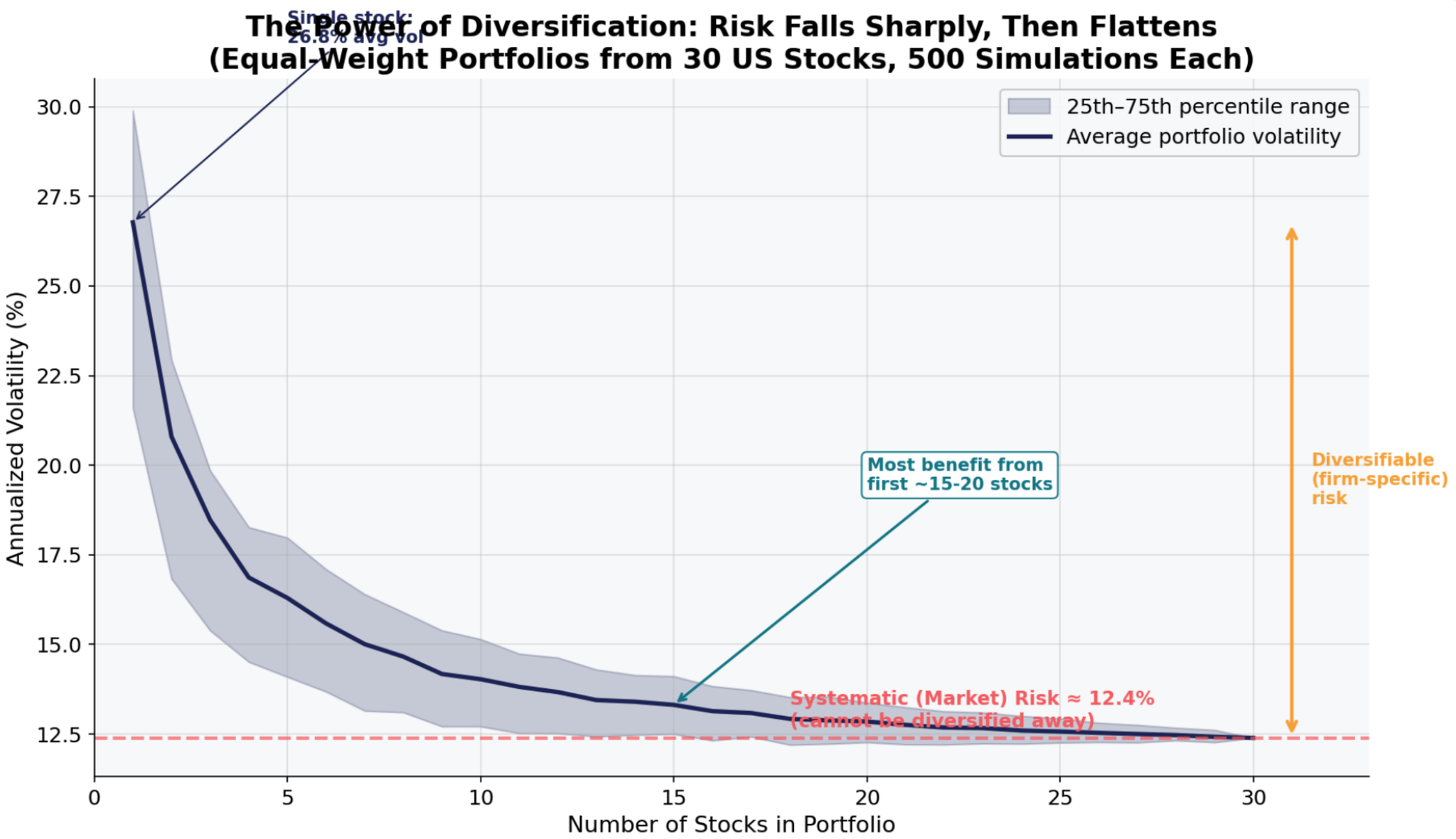
To avoid high drawdowns (portfolio decreases) you should trade stocks that are less correlated

Portfolio Construction and The Efficient Frontier



- Each dot is one of 15,000 randomly generated portfolios, with a different combination of the 6 assets
- Investors want high return and low vol
- Upper left edge is the efficient frontier
- Any dot on the inside is inferior
- Two critical points: min risk point, max sharp point
- Note portfolio diversification moves you to the frontier!

How Much Diversification? 15-20 Stocks is Fine



Connecting the Concepts

Marginal Buyer/Seller	Prices form through real-time auction between most motivated traders	Order book modeling, high-frequency trading algorithms
Return Distributions	Fat tails mean extreme events more common than normal distribution predicts	Better risk models accounting for tail risk
CAPM / Beta	Expected return based on systematic risk exposure	Factor models, alpha generation strategies
Options	Asymmetric risk-reward through limited downside, unlimited upside	Options pricing, portfolio hedging, volatility trading
Diversification	Combining uncorrelated assets reduces total portfolio risk	Portfolio optimization, asset allocation

AI doesn't replace this theoretical foundation—it builds upon it. Machine learning excels at finding patterns in high-dimensional data, but understanding *why* patterns exist requires financial theory. The most powerful applications combine both.

Today's Lab Preview

1. **Distribution Testing**
   Test whether real stock returns follow normal distributions using histograms, QQ plots, and the Jarque-Bera statistical test

2. **CAPM Regression**
   Calculate beta and alpha for selected stocks using regression analysis against market returns

3. **Option Payoffs**
   Plot payoff diagrams for call and put options at expiration across different strike prices

4. **Options Chain Analysis**
   Explore real options chain data from Yahoo Finance, examining bid-ask spreads and implied volatility

5. **Portfolio Construction**
   Build a simple two-stock portfolio and observe diversification benefits through correlation analysis

6. **AI-Assisted Interpretation**
   Practice prompting generative AI to explain your analytical results and suggest next steps

Key Takeaways

Price Formation

Stock prices are set by the marginal buyer and seller—not the average opinion. Understanding this microstructure is essential for AI trading strategies.

CAPM Framework

Expected return = risk-free rate + beta × market premium. This relationship provides the foundation for evaluating investment performance.

Options Power

Options create asymmetric payoffs—limited downside with substantial upside potential. This makes them powerful for speculation and hedging.

Fat Tails Matter

Returns exhibit fat tails—extreme events occur far more frequently than normal distribution predicts. This fundamentally affects risk management and model design.

Beta and Alpha

Beta measures systematic risk exposure. Alpha measures outperformance relative to risk—the holy grail of active management.

Diversification Benefit

Combining uncorrelated assets reduces portfolio risk below the weighted average of individual risks—the foundation of portfolio theory.

Preparing for Next Week

Required Reading

Check Canvas for next week's assigned readings. We'll build on today's foundation with more advanced portfolio theory and factor models.

Discussion Questions

Come prepared to discuss these provocative questions:

- Why do most active managers fail to generate positive alpha consistently?
- How might AI and machine learning change the efficient market hypothesis?
- Can markets be both informationally efficient and exhibit persistent anomalies?

Next Week's Focus

We'll take a deeper dive into data-driven finance, exploring how to extract signals from market data, backtest trading strategies systematically, and evaluate performance rigorously.

The lab will introduce more advanced Python libraries for financial analysis and walk through a complete strategy development workflow from hypothesis to implementation.

This progression builds toward your final projects—developing original AI-powered trading or risk management applications.

Questions and Discussion

Office Hours

Wednesdays 6:00-7:00 PM (right before class)—come with questions about concepts, lab exercises, or project ideas

Contact Information

Email: hersh@chapman.edu for scheduling individual meetings or technical assistance

Course Materials

All lecture slides, lab notebooks, datasets, and supplementary resources available on Canvas

> "The stock market is filled with individuals who know the price of everything, but the value of nothing." — Philip Fisher

Let's head to the lab and put these concepts into practice!

Additional Resources

Recommended Textbooks

- Bodie, Kane, Marcus: *Investments* (comprehensive coverage of CAPM and portfolio theory)
- Hull: *Options, Futures, and Other Derivatives* (definitive options reference)
- Lopez de Prado: *Advances in Financial Machine Learning* (AI applications in finance)

Online Resources

- Quantopian Lectures (archived): excellent tutorials on quantitative finance
- QuantConnect: platform for algorithmic trading strategy development
- SSRN Financial Economics papers: cutting-edge academic research
- ArXiv quantitative finance section: latest preprints and working papers

$1T

Daily Options Volume

Notional value traded in options markets daily globally

50+

Years of CAPM

The Capital Asset Pricing Model remains central after five decades

5-10%

Alpha Generators

Percentage of active managers consistently outperforming benchmarks long-term